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FUM DCF V1
Radio clock module DCF -**Operation / Software**

1. General

The radio clock module is a very small circuit board module whose primary application is the reception and decoding of the DCF77 time signal. The decoded and verified time information forms the internal time base for the module and is provided for further processing at module output X4 (RS232-OUT). This radio clock module communicates with any connected host computer via a serial interface. The command set described in section 3.4 is used for this purpose. This provides the connected host computer with fully verified time and status information at all times via the serial interface.

The radio clock module automatically initiates radio receptions for synchronization with the DCF77 signal several times a day. Additional reception attempts can be initiated either manually (operating button, see section 4) or upon request from the connected host computer (via serial interface; sections 3.4.5 and 3.4.6).
For successful use, the radio clock module must be connected to a tuned antenna, the serial interface of a host computer (with appropriate level adjustment) and the supply voltage.

2. Radio reception

After applying the operating voltage (X1: +3V; X2: GND) and running the test routine, the radio clock module immediately begins radio reception in order to have valid time information available as quickly as possible. This initial reception takes approximately 3 minutes under interference-free reception conditions. The successfully decoded, complete time information then serves as the basis for the module's internal time base. After the first successful reception, further, automatically initiated reception attempts take place at 8:45 p.m., 9:45 p.m., etc., every hour until 3:45 a.m. Additional reception attempts (if necessary) can be initiated via the serial interface (see section 3.4.5. or 3.4.6.) or using the "button" control input (see section 4.). The duration of the reception attempts is limited to 10 minutes each; with good reception, each attempt lasts 65 seconds.

The received time telegrams are subjected to a three-fold check (plausibility, parity, etc.). Only when all conditions are met is the corresponding time telegram transferred to the time register of the radio clock module.

Notice :

After applying the operating voltage, the time registers of the radio clock module function as a type of counter, starting at 0:00 until the register contents are updated by the data of a verified time telegram. Thus, correct, analyzable time information is only available at the serial interface after a successful DCF77 reception has occurred.

If an initial reception fails after the module is put into operation, a new reception attempt is automatically initiated every hour at the 45th minute of the internal time base until a successful reception is achieved. The module then enters the reception mode described above.

3. The serial interface

3.1 General

The serial interface consists of one line as a data input (X3) and one line as a data output (X4). This allows data to be requested from the radio clock module via the serial interface and, to a limited extent, also to control the radio clock module. Data traffic on both lines occurs at 300 bps. Due to the limited resources of the radio clock module, a simple software protocol is used. Handshake lines are not available. However, simple synchronization is possible based on the character echo.

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3.2. Data format

3.2.1. Data or commands for the radio clock module

3.2.1.1. Interface parameters

	Compatibility mode
Speed:	300 bps
Data bits:	8
Parity bit:	no
Stop bits:	2

3.2.1.2. Data entry function

The radio clock module accepts commands that, including their parameters, can consist of up to three characters. A subsequent "CR" character (hex 0d) causes the radio clock module to execute the command. A pause of approximately 100ms should be allowed between consecutive characters or between the command character and the "CR". The command characters and their parameters are only decoded in the four least significant bits. This means that "?", "O", and "o", for example, always have the same effect (hex 3f, 4f, and 6f). Only the "CR" that concludes the command is fully decoded. Internally, there is a FIFO buffer for the last four characters received. When a command is entered, the parameters (one or two characters) are sent first (if necessary), followed by the command identifier (one character). All received characters are stored in the FIFO buffer. If a "CR" is found at the end of this buffer, the three preceding characters are interpreted as a command. For commands that require parameters, these must not be omitted, as they are taken from the otherwise undefined first two positions of the FIFO buffer.

Danger:
Due to incomplete decoding of the command characters, errors can easily occur when testing communication with a simple terminal program, for example. This could result in misinterpretation of the commands or undefined register settings.

3.2.2. Responses of the radio clock module

3.2.2.1. Interface parameters

Speed:	300 bps
Data bits:	7
Parity bit:	straight
Stop bits:	2

3.2.2.2. Function of data output

A response from the radio clock module consists of a sequence of ASCII characters and a terminating character “CR”.

The range of values is limited to the digits 0 ... 9 and the characters : ; < = > ?. The information is always contained in the four least significant bits.
Bits 4 and 5 of each responded character are always set to one, bit 6 is set to zero, and bit 7 is always set according to the even parity of the character. This results in the 16 characters of the value set mentioned above.

During the response of the radio clock module there is no possibility to interrupt the serial output, ie all characters must be received immediately on the other side.

3.2.3. Interface parameters on standard interfaces

Due to the different parameters at data input (8 bits, no parity) and data output (7 bits, even parity), the following procedure is recommended:

a) If data is to be displayed immediately (e.g. with a simple terminal program such as “Windows Terminal”), 7 data bits and the parity bit should be set to “space”.

b) If data is evaluated via software, 8 data bits and no parity should be set. The parity bit must then be evaluated or masked by the host computer via software.

3.2.4 Support for simplified data exchange

In order to keep the software effort for host computers without a serial interface as low as possible, a simplified form of data exchange is possible.

The command in 3.4.1. (Send time log) can be triggered more easily than using a sequence of ASCII characters. To do this, the level on the receive line (X3) of the radio clock module must be kept active until the time log has begun to be transmitted on the transmit side (X4). This can take a maximum of one second, as the first start bit of the time log is always transmitted on the full second and thus marks the beginning of the second. Line X3 is normally high (inactive) and must be pulled low (active) to transmit the time log. This corresponds to the inverted level of the RS232 interface.

The level of input line X3 can be kept active for any length of time, so that, for example, the time protocol is continuously transmitted every full second. However, with this mode of operation, it should be noted that the current consumption of the radio clock module increases to approximately 200µA during the approximately 600ms time protocol transmission.

3.3. Handshaking

The radio clock module immediately returns each received character as an echo. The next character may only be sent to the radio clock module after the previous character has been received as an echo and an additional wait time of 5 ms has been observed. The echo is always unconditional, i.e., regardless of whether it can be interpreted as a command. The only condition is the correct occurrence of the start bit and the two stop bits. Bit 7 of the echo character (in the responses from the radio clock module, this is usually the parity bit) corresponds to that of the received character and is not changed in the radio clock module.

3.4. Description of the commands

3.4.1. Send time log

ASCII syntax: o <CR>
Instead of “o” you can use any other character whose code ends with f(hex), e.g. / ? O _

Binary syntax: xxxx1111 00001101

The radio clock module responds to this command with 15 charactersand a final <CR>. These 15 characters contain the complete time information.
The response is not immediate, but only at the beginning of the next second, which is marked by the first edge of the start bit of the first character (hour tens).
The individual characters have the following meaning:

1. Hourly tenner
2. Hour one
3. Minute tens
4. Minute one
5. Second tens
6. Second One
7. Weekday 1 (Monday) ... 7 (Sunday)
8. Daily Tenner
9. Day One

10th monthly tenner
11th month
12th decade
13th annual

14. Bits 16 ... 19 of the DCF77 protocol

Bit7	parity	
Bit6	always 0	
Bit5	always 1	
Bit4	always 1	
Bit3	Announcement leap second =1 during CET, =0	(Bit 19 of the DCF77 protocol)
Bit2	during CEST =0 during CET, =1 during CEST	(Bit 18 of the DCF77 protocol)
Bit1	Announcement change CET-CEST and vice versa	(Bit 17 of the DCF77 protocol)
Bit0		(Bit16 of the DCF77 protocol)

15. Status

Bit7	parity
Bit6	always 0
Bit5	always 1
Bit4	always 1
Bit3	=1 if battery voltage is too low
Bit2	=1 if a radio reception was aborted unsuccessfullyand no valid time information is available yet. This can occur after a reset under poor reception conditions. This bit is reset upon the first successful reception and then remains 0.
Bit1	=1 if the previous reception attempt was successful =0 if the previous reception attempt was unsuccessful =1 if valid time
Bit0	information is present This bit is =0 after reset and is set with the first successful reception attempt.

3.4.2. Send time log world time

ASCII syntax: e <CR>
Instead of “e” all other characters are possible whose code ends with 5(hex)

Binary syntax: xxxx0101 00001101

The radio clock module responds to this command with 15 charactersand a final <CR>. These 15 characters contain the complete time information.
The response is not immediate, but only at the beginning of the next second, which is marked by the first edge of the start bit of the first character (hour tens).

The individual characters have the following meaning:

1. Hourly tenner
2. Hour one
3. Minute tens
4. Minute one
5. Second tens
6. Second One
7. Weekday 1 (Monday) ... 7 (Sunday)
8. Daily Tenner
9. Day One
10th monthly tenner
11th month
12th decade
13th annual
14. Bits 16 ... 19 of the DCF77 protocol

Bit7	parity	
Bit6	always 0	
Bit5	always 1	
Bit4	always 1	
Bit3	Announcement leap second =1 during CET, =0	(Bit 19 of the DCF77 protocol)
Bit2	during CEST =0 during CET, =1 during CEST	(Bit 18 of the DCF77 protocol)
Bit1	Announcement change CET-CEST and vice versa	(Bit 17 of the DCF77 protocol)
Bit0		(Bit16 of the DCF77 protocol)

15. Status

Bit7	parity
Bit6	always 0
Bit5	always 1
Bit4	always 1
Bit3	=1 if battery voltage is too low
Bit2	=1 if a radio reception was aborted unsuccessfully and no valid time information is available yet. This can occur after a reset under poor reception conditions. This bit is reset upon the first successful reception and then remains 0.
Bit1	=1 if the previous reception attempt was successful =0 if the previous reception attempt was unsuccessful =1 if valid time
Bit0	information is present This bit is =0 after reset and is set with the first successful reception attempt.

3.4.3. Send operating mode

ASCII syntax: f <CR>
Instead of “f” all other characters are possible whose code ends with 6(hex)

Binary syntax: xxxx0110 00001101

The radio clock module responds to this command with 4 characters and a final <CR>. The four characters have the following meaning:

1. Tens of the number of hours since the last successful reception
2. One of the number of hours since the last successful reception
3. Status information

Bit7	parity
Bit6	always 0
Bit5	always 1
Bit4	always 1
Bit3	=1 at DCF77 =0 at MSF
Bit2	=1 if parity and DCF77 in second 59 compatible with the old version (compatibility mode) without
Bit1	meaning
Bit0	Status of the switching output

4. ASCII “0” character

3.4.4. Send radio reception and line status

ASCII syntax: g <CR>
Instead of “g” all other characters are possible whose code ends with 7, for example ' 7 GW w

Binary syntax: xxxx0111 00001101

The radio clock module responds to this command with 2 charactersand a final <CR>. The two characters have the following meaning:

1. Radio reception and line status Bit7
parity
Bit6 always 0
Bit5 always 1
Bit4 always 1
Bit3 always 0
Bit2 always 0
Bit1 =0 at low level on controller pin IP51 =1
at high level on controller pin IP51
Bit0 =0 if no radio reception attempt is currently taking place =1
if a radio reception attempt is currently taking place

2. Radio reception quality index in the range 0 ... 5 5
corresponds to completely undisturbed reception
0 corresponds to very poor reception
At 0, 1, and 2, reception is usually not expected. In the interval between reception attempts, the quality index is always 0.

3.4.5. Immediately perform radio reception

ASCII syntax: h <CR>
Instead of “h” all other characters are possible whose code ends with 8, for example (8 HX x

Binary syntax: xxxx1000 00001101

Following this command, the radio clock module switches on the receiver and performs a reception attempt, which includes reading the complete DCF77 time information and synchronizing (accurate to 20 ms) the radio clock module's time grid. If this attempt is unsuccessful, bit 1 of the 15th character of the time protocol (see 3.4.1 or 3.4.2) is set to zero. If the reception attempt is successful, this bit is always set.

3.4.6. Immediately perform radio reception

ASCII syntax: i <CR>
Instead of “i” you can use any other character whose code ends with 9, for example) 9 IY y

Binary syntax: xxxx1001 00001101

This command has the same function as described in 3.4.5 and is included for compatibility reasons.

4. Control connection “button”

A radio reception attempt can be initiated at any time via this connection (via a button or via the control line of the connected host computer). The reception attempt is triggered by the high/low edge of the applied control pulse.

The LP1 terminal has an internal 300 kOhm pull-up resistor. To activate the receive function, LP1 is connected to GND for at least 125 ms. Theoretically, LP1 can be held at GND for any length of time, but it should be noted that as long as LP1 is connected to GND, no time information is transferred to the radio clock registers. Therefore, LP1 should normally be high.

Subject to change